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A Decision Framework for CO2 Storage Site Selection in Depleted Reservoirs

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Abstract

As Malaysia pushes toward its national net-zero emissions goal, carbon capture and storage (CCS) is emerging as a key decarbonization strategy for the oil and gas sector. Depleted hydrocarbon reservoirs, with their well-understood geology and existing infrastructure, offer promising sites for permanent CO_2 storage. This study introduces a decision-support framework designed to optimize CO_2 injection site selection across Malaysian reservoirs, aiming to maximize storage potential while maintaining pressure safety and reducing logistical hurdles. The framework is specifically tailored for early-stage CCS project screening and planning.

We developed a mixed-integer linear programming (MILP) model to identify the best reservoirs for CO_2 injection. Each candidate reservoir was evaluated based on storage capacity, distance to CO_2 compression platforms, and pressure response under injection. A simplified gas tank model estimates pressure buildup from CO_2 injection, with constraints to avoid exceeding fracture pressure or operational safety limits. The model balances CO_2 storage volume against transport costs linked to source-to-sink distance, maximizing net storage utility. Additionally, we also explored and compared alternative nature-inspired optimization algorithms as part of the study.

When applied to real-world Malaysian reservoir data, the framework revealed that pressure constraints and location-based costs heavily influence injection site selection. Some high-capacity reservoirs were deprioritized due to pressure risks or unfavorable distances. The MILP model allows rapid scenario testing and sensitivity analysis, highlighting the need for early pressure screening to prevent costly CO_2 allocation mistakes in later project stages.

This study presents a practical optimization approach for Malaysia's regional CCS planning, supporting industry efforts to achieve net-zero targets. By integrating reservoir capacity, geographic logistics, and pressure safety constraints, the framework offers a scalable and efficient alternative to traditional screening methods. The inclusion of a gas-law-based pressure model at the pre-screening stage ensures realistic and safe site evaluations without extensive simulations.

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Introduction

Currently, PETRONAS is spearheading large-scale CCS initiatives to establish Malaysia as a regional storage hub, with several candidate storage locations identified, including the Duyong, Penyu, and Lawit fields offshore Peninsular Malaysia (PETRONAS 2025). While depleted hydrocarbon reservoirs offer favorable characteristics such as known geology and existing infrastructure, selecting suitable storage sites remains technically complex and requires careful consideration of pressure management and geomechanical constraints to ensure long-term storage integrity (Mustafa et al. 2022).

Central to the success of such large-scale deployment is the ability to model and optimize the CO_2 supply chain from capture to storage. Transporting CO_2 at pressures above the critical point is generally advantageous because dense-phase CO_2 exhibits high density, which reduces volumetric flow rates and can lower pipeline diameters and compression requirements. However, operation in this regime introduces important technical challenges. The thermophysical properties of CO_2 near the critical region are highly sensitive to pressure, temperature, and fluid composition. Small concentrations of impurities such as hydrogen, nitrogen, or methane can significantly modify phase behavior, density, viscosity, and compressibility, thereby affecting hydraulic performance, compression power requirements, and flow assurance margins. Previous studies have shown that CO_2 purity can influence pipeline design, operating pressure windows, and overall transport network cost, particularly in shared infrastructure serving multiple capture sources (Wetenhall et al. 2014). As a result, accounting for these sensitivities is essential when evaluating dense-phase CO_2 transport and storage options at an early stage.

Research has shown that hub-based optimization models can significantly reduce CCS infrastructure costs through economies of scale and centralized transport and storage systems (Zhang et al. 2018). However, many traditional optimization frameworks treat storage reservoirs as static sinks with fixed capacity limits. More recent studies demonstrate that integrating physics-based reservoir dynamics, including pressure evolution and injectivity constraints, directly into mixed-integer linear programming (MILP) formulations provides a more realistic assessment of both transport feasibility and storage performance (D'Amore and Bezzo 2021; Bertoni et al. 2025).

Preliminary studies using nodal analysis further highlight the value of evaluating CO_2 injectivity and its interaction with surface and subsurface infrastructure at an early stage of the project lifecycle (An and Kwon 2025). Such early-stage assessments are particularly important given the strong thermodynamic sensitivities of dense-phase CO_2, which can complicate pipeline hydraulics and flow assurance. Although detailed numerical simulations are typically deferred until the Front-End Engineering Design (FEED) stage, previous work indicates that integral thermo-hydraulic models can provide sufficiently reliable predictions of dense-phase CO_2 transport for initial screening and concept selection (Martynov et al. 2015).

CCS Value Chain

The transport of CO_2 molecule in a CCS chain spans hundreds of kilometers and several kilometers of vertical depth. Unlike natural gas or oil, CO_2 exhibits a unique sensitivity to pressure and temperature near its critical point (73.8 bar and 31.1°C), making the understanding of the expected pressure and temperatures profile a fundamental requirement for both design and operation.

Figure 1 illustrates the main components of a CCS infrastructure. The system begins at the capture site, which may include facilities such as cement plants, steel mills, or waste-to-energy plants. The CO_2 stream exits the capture process at pressures that depend on the specific capture technology. In post-combustion amine-based systems, CO_2 is typically recovered at near-atmospheric pressures, on the order of 1–3 bar, whereas pre-combustion or membrane-based systems may deliver CO_2 at higher pressures. In some cases, the capture site may also incorporate compression facilities, as is common for existing oil and gas production installations processing high-CO_2 gas streams. Alternatively, CO_2 may be transported in liquid form by

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ship from remote sources and delivered to a central hub for pumping and injection, as demonstrated by the Northern Lights project in Norway.

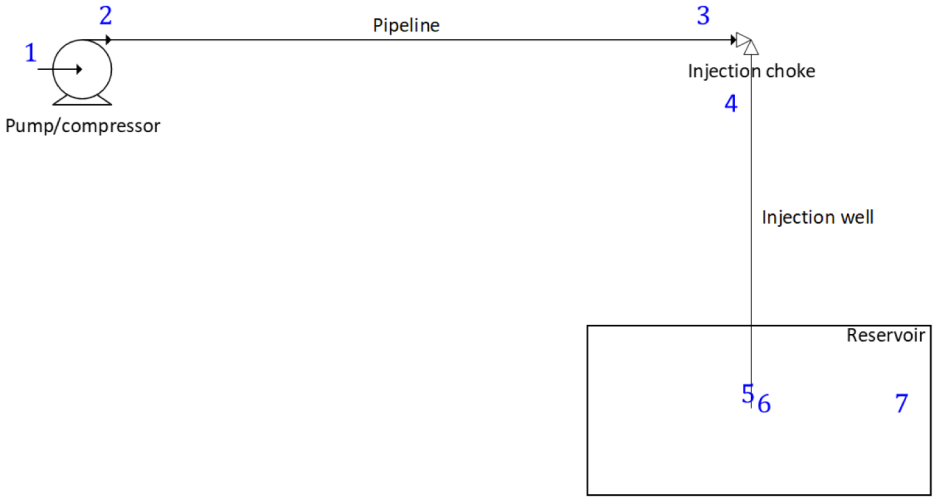


Figure 1—Typical CCS value chain, from pumping or compression to the reservoir for storage

Figure 2 illustrates the corresponding pressure profile from Point 1 to Point 7.

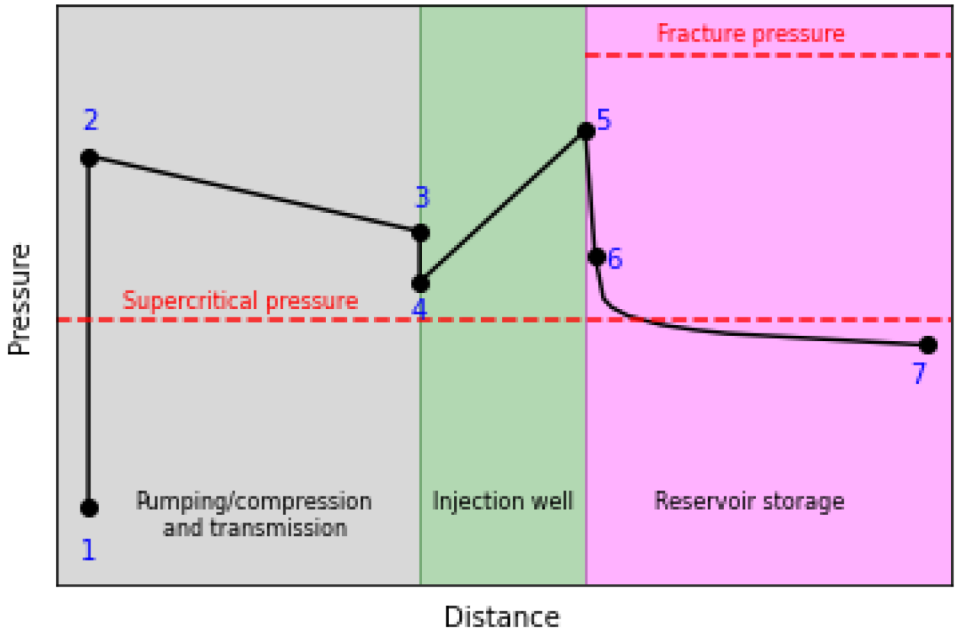


Figure 2—Typical pressure profile of typical CCS design from the CO2 source to the storage reservoir

Point 1: The stream at this stage is a low-density gas, highly sensitive to impurities. The presence of non-condensable gases such as nitrogen and hydrogen significantly alters compression or pumping requirements. These "light" components increase the work required by compressors and shift the phase envelope, often raising the cricondenbar of the mixture, which necessitates higher pressurization than pure CO_2 to ensure a stable dense-phase transition.

Point 2: At the discharge of the compression or pumping train, the CO_2 is converted into a dense fluid. This state is referred to as supercritical when both the critical pressure and temperature are exceeded, and as dense phase when the pressure exceeds the critical pressure but the temperature remains lower, as is

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typical in subsea environments. Target delivery pressures are commonly in the range of 100-200 bar to overcome cumulative frictional losses and reservoir backpressure while maintaining an adequate margin above the two-phase region. Operation in the dense phase combines gas-like viscosity with liquid-like density, typically in the range of 600-900 kg/m^3, enabling high mass flow rates through the pipeline and into the reservoir.

Point 2 to 3: Along the CO_2 transmission segment, the pressure profile is governed by Darcy—Weisbach frictional losses and hydrostatic effects associated with seafloor bathymetry. Dense-phase transport is hydraulically efficient, and large trunklines, typically 16–24 inches in diameter, may exhibit pressure drops on the order of 0.1 bar per kilometer, depending on flow rate and fluid density. Subsea pipelines may approach thermal equilibrium with the seabed over hundred meter-scale distances, although the rate of convergence depends on insulation and operating conditions. Seabed temperatures are commonly in the range of 20–30°C in tropical regions and may decrease to approximately 4°C in cold deepwater. Lower temperatures increase CO_2 density, reducing flow velocity for a given mass rate and lowering frictional pressure losses. However, excessive pressure reduction may lead to entry into the two-phase region, resulting in slug flow, mechanical vibration, and unstable wellhead conditions.

Point 3 to 4: The transition from pipeline to wellbore is controlled by the injection choke, which serves as the primary point of active operational control. The choke dissipates excess pipeline pressure and regulates the injection rate. As CO_2 passes through the choke, it undergoes near-isenthalpic expansion. Because CO_2 exhibits a relatively high Joule—Thomson coefficient under these conditions, significant throttling can lead to rapid temperature reduction. Very low CO_2 temperatures, below approximately –20°C, pose risks of thermal shock to steel, seal embrittlement, and the formation of hydrates or solid CO_2.

Point 4 to 5: As CO_2 descends through the injection tubing, the pressure profile is dominated by the hydrostatic head of the fluid column. In deep offshore wells, the weight of a dense-phase CO_2 column can contribute tens of bar, and in very deep wells potentially approaching hundreds of bar, to the pressure available at the sandface, in some cases enabling near-passive injection. Geothermal warming during descent increases fluid temperature, which reduces CO_2 density and correspondingly lowers the hydrostatic contribution.

Point 5 to 6: At the sandface, injection occurs when the flowing bottomhole pressure P_{bh} exceeds the reservoir pressure P_r. Although the damaged zone around the wellbore, arising from drilling, fines migration, or scale deposition, is physically limited in extent, it can impose substantial additional pressure loss at CCS-scale injection rates. Skin represents a localized resistance that effectively consumes available sandface pressure. Even moderate positive skin values can reduce injectivity by tens of percent and require additional pressure on the order of tens of bar.

Point 6 to 7: At the formation scale, reservoir pressure evolution is governed by cumulative injected mass or volume. For early-stage assessment, the P/z tank model provides a suitable representation, linking cumulative injection to average reservoir pressure through a linear relationship. This temporal coupling is important when optimization must balance short-term injection targets against longer-term pressure build-up and storage constraints.

Methods

With the end-to-end pressure profile established, the analysis shifts to the hydraulic constraints used for reservoir selection within a CCS hub. These calculations enable rapid screening of candidate sites during the early project phases, by identifying injection-rate and pressure limitations that may constrain feasibility. By applying hydraulic limits upfront, potential bottlenecks can be identified before undertaking more detailed and capital-intensive studies. Geomechanical and geochemical effects, while critical to long-term storage integrity, are beyond the scope of this work and are considered separately.

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Overview of the modelling framework

A dynamic, pressure-aware optimisation framework is proposed. The framework couples:

1. A physics-based transport and injection model, which evaluates the cost and pressure response associated with injecting CO_2 into a given reservoir at its current pressure state; and
2. A linear optimisation layer, which allocates the required CO_2 disposal rate across candidate reservoirs on an annual basis.

The framework is designed explicitly for screening and pre-FEED analysis, where rapid scenario evaluation and transparent physical interpretation are prioritised over detailed geomechanical or multiphase simulation.

Transport and Injection Physics

Pipeline hydraulics. CO_2 transport is assumed to occur in the dense phase to minimize volumetric flow rates and associated compression requirements. For a pipeline of internal diameter D, length L, mass flow rate $\dot{m}$, and fluid density ρ, the frictional pressure gradient along the pipeline is computed using the Darcy–Weisbach formulation:

$$\frac{dP}{dx} = \frac{f}{2D}\rho v^2 \tag{1}$$

where P is the pressure, $x$is the axial coordinate along the pipeline, v is the average flow velocity, and f is the Darcy friction factor. The average velocity is given by

$$v = \frac{\dot{m}}{\rho A}$$

where $\dot{m}$is the mass flow rate and $A = \pi D^2/4$ is the internal cross-sectional area of the pipeline.

The Darcy friction factor f is evaluated using an explicit correlation applicable to fully turbulent flow in rough pipes:

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\varepsilon}{3.7D} + \frac{5.74}{\mathrm{Re}^{0.9}}\right)\right]^2} \tag{2}$$

where $\varepsilon$is the absolute pipe roughness, D is the internal pipe diameter, and Re is the Reynolds number defined as $\mathrm{Re} = \rho v D/\mu$, with μ denoting the dynamic viscosity of CO_2. This correlation provides a computationally efficient approximation to the Colebrook–White equation and is suitable for screening-level analysis of dense-phase CO_2 transport under turbulent flow conditions.

Reservoir pressure build-up (tank model). The P/z tank model is used to represent reservoir pressure evolution because it provides a simple linear relationship between cumulative injected mass and average reservoir pressure for preliminary screening. However, the P/z approximation assumes spatially uniform pressure and single-phase compressible behavior; it does not capture near-well pressure gradients, multiphase effects, or the strong pressure–temperature-dependent compressibility of near-critical CO_2. Consequently, the P/z model is treated as a conservative screening tool.

In this method, the reservoir pressure increment associated with cumulative injected mass is given by:

$$\frac{P_r^{t+1}}{z\left(P_r^{t+1}\right)} = \frac{P_r^t}{z\left(P_r^t\right)} + \frac{\Delta m_t RT}{MV_p} \tag{3}$$

where P_r is the average reservoir pressure, $z$is the gas compressibility factor evaluated at reservoir conditions, Δm_t is the injected mass increment at time step t, R is the universal gas constant, T is reservoir temperature, M is the molar mass of CO_2, and V_p is the effective pore volume.

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A similar P/z-based scaling has been used in previous studies of CO_2 storage in depleted gas reservoirs. For example, Akai et al. (2021) relate the equivalent storage volume of CO_2 to the original methane gas volume using the ratio of compressibility factors evaluated at reservoir conditions, and validate this assumption against numerical reservoir simulation results.

It is noted that for storage formations containing low-compressibility fluids, such as saline aquifers, a direct mass balance based on fluid and rock compressibility may provide a more appropriate representation of pressure evolution than the P/z approach. The P/z model is therefore most suitable for gas-dominated or highly compressible systems, and its application here is intended solely for preliminary screening rather than detailed reservoir performance prediction.

Storage integrity is enforced by imposing an upper bound on reservoir pressure:

$$P_r \le P_{frac} \tag{4}$$

where P_{frac} is the fracture pressure of the storage formation. Once this limit is reached, the reservoir is removed from further injection. In principle, fracture risk is governed by near-wellbore pressure; therefore, a more stringent constraint of $P_{bh} \le P_{frac}$ may be applied in detailed well-scale analyses. In the present screening-level model, the reservoir pressure constraint is adopted as a conservative proxy to ensure storage integrity.

This limit enforces storage capacity through pressure evolution rather than through volumetric constraints imposed independently of pressure.

Well injectivity. The flowing bottom-hole pressure required to inject CO_2 at a mass rate $\dot{m}$ into a reservoir with average pressure P_r is calculated using the radial Darcy flow equation for compressible fluids (Dake, 1978):

$$P_{bh}^2 = P_r^2 + \frac{\dot{m}\mu z R T}{M k h}\left[\ln\left(\frac{r_e}{r_w}\right) - 0.75\right] \tag{5}$$

where P_{bh} is the flowing bottom-hole pressure, $\dot{m}$ is the CO_2 mass injection rate, μ is the CO_2 viscosity at reservoir conditions, z is the gas compressibility factor, $R$is the universal gas constant, T is the reservoir temperature, M is the molar mass of CO_2, k is the reservoir permeability, h is the reservoir thickness, r_e is the effective drainage radius, and r_W is the wellbore radius. Skin effects are intentionally neglected at this stage to preserve a first-order screening representation of injectivity, with the understanding that near-wellbore effects will be addressed in subsequent detailed well-scale analyses.

Compression power and operating cost. The total pressure rise (ΔP_{tot}) required from the capture facility (P_{cap}) to the reservoir is:

$$\Delta P_{\text{tot}} = \left(P_{bh} + \Delta P_{\text{pipe}}\right) - P_{\text{cap}} \tag{6}$$

The corresponding compressor shaft power (W) is approximated as:

$$W = \frac{\dot{m}}{\rho}\frac{\Delta P_{\text{tot}}}{\eta} \tag{7}$$

leading to an annual operating cost:

$$C_{OPEX} = W \times 8760 \times C_{el} \tag{8}$$

Here, the operating cost (C_{OPEX}) is proportional to power and electricity cost (C_{el}).

Pipeline capital cost. Pipeline capital cost (C_{CAPEX}) is represented using a screening-level power-law relation (McCoy and Rubin, 2008):

$$C_{\text{CAPEX}} = k_{\text{capex}} D^{1.3} L \tag{9}$$

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Here, k_{capex} is the multiplier to adjust the empirical capital cost, with a value of 1.2 was used in this study. For optimization purposes, CAPEX is annualized over the project duration. For cost accounting and interpretation, CAPEX is treated as a one-time investment incurred in the year a pipeline is first utilized.

Single-reservoir cost minimization

For each reservoir and each pressure state, the pipeline diameter is optimised by solving:

$$\min_{D}\left(C_{\mathrm{CAPEX,ann}}(D)+C_{\mathrm{OPEX}}(D)\right) \tag{10}$$

This is a smooth one-dimensional nonlinear problem. The resulting marginal cost represents the cost of injecting an incremental unit of CO_2 into that reservoir at its current pressure.

Dynamic allocation model

Let $q_{i,t}$ denote the injection rate into reservoir i in year t, and $Q_t$the total injection requirement.

For each year, the optimisation solves:

$$\min_{q_{i,t}} \sum_{i} c_i\left(P_{r,i}^{t}\right) q_{i,t} \tag{11}$$

subject to:

$$\sum_{i} q_{i,t} = Q_t \tag{12}$$

$$0 \le q_{i,t} \le q_{i,\max} \tag{13}$$

Reservoir pressures are updated after each year using Eq. (3), and reservoirs are closed irreversibly once Eq. (4) is reached.

Although the framework is structured to accommodate a full MILP with binary activation variables, the present implementation focuses on economic dispatch with pressure-limited availability. For brevity, the MILP is not discussed in the paper and instead the focus is given on how the realistic hydraulics is incorporated.

Case study: Hub-and-spoke distribution

All computations in this study were performed using open-source libraries implemented in Python, including NumPy for numerical calculations (Harris et al., 2020), PuLP for linear optimization (Mitchell et al., 2011), ThermoPack and CoolProp for fluid property functions (Wilhelmsen et al., 2017; Bell *et al*., 2013). To protect commercially sensitive and region-specific information, the physical, operational, and economic input data used in the analysis have been modified and anonymized.

To demonstrate the proposed framework, a CCS hub supplying up to 10 Mtpa of captured CO_2 over a 20-year project life is analyzed. The injection demand ramps up over the first five years of operation and then remains constant for the remainder of the project.

Four candidate storage reservoirs are considered, comprising one near-field, two mid-distance fields, and one far-field, as illustrated in Figure 3. The reservoirs differ in distance from the capture hub, initial pressure, injectivity, and effective storage capacity. A hub-and-spoke transport configuration is adopted as a baseline case because of its conceptual simplicity. Under this configuration, each reservoir is connected to the hub by an individual pipeline.

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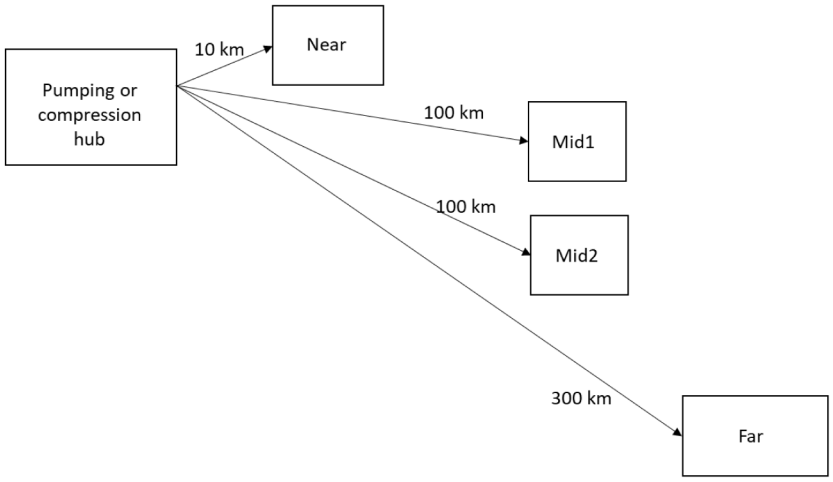


Figure 3—CCS hub to reservoirs configuration

Dense-phase CO_2 transport and injection are assumed throughout the analysis.

Results and observations

Figure 4 shows the annual CO_2 injection allocation across candidate reservoirs where the dashed line gives the total injection volumes. Injection transitions from near-field to far-field storage as pressure constraints tighten and marginal costs evolve.

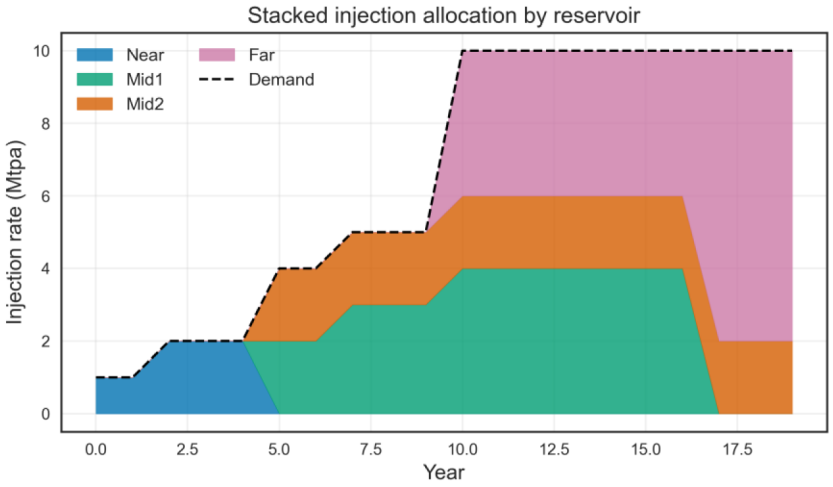


Figure 4—Annual injection allocation by reservoir

Figure 5 presents the annual CCS system cost, decomposed into pipeline capital expenditure (CAPEX) and operating expenditure (OPEX). Pipeline CAPEX appears as discrete spikes in the year a storage site is first utilized, reflecting the assumption that pipeline infrastructure is constructed on a just-in-time basis when injection begins. In contrast, operating costs scale with injection rate and persist throughout the project lifetime.

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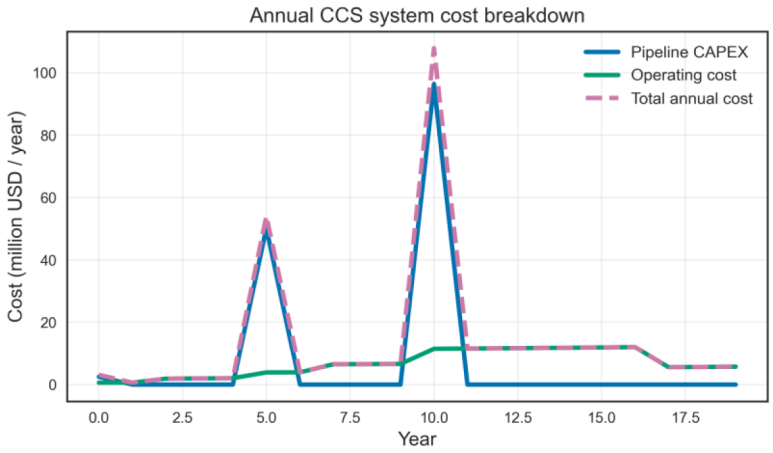


Figure 5—Annual CCS system cost

Although the far-field pipeline incurs a relatively large upfront investment, its superior injectivity and pressure performance result in lower operating costs in subsequent years. This highlights the trade-off between early infrastructure investment and long-term operating efficiency.

Figure 6 shows the cumulative cost profile of a CCS system over the project lifetime, highlighting step changes around Years 5 and 10 corresponding to the activation of additional pipeline infrastructure, followed by more gradual cost increases contributed by the operating expenditure.

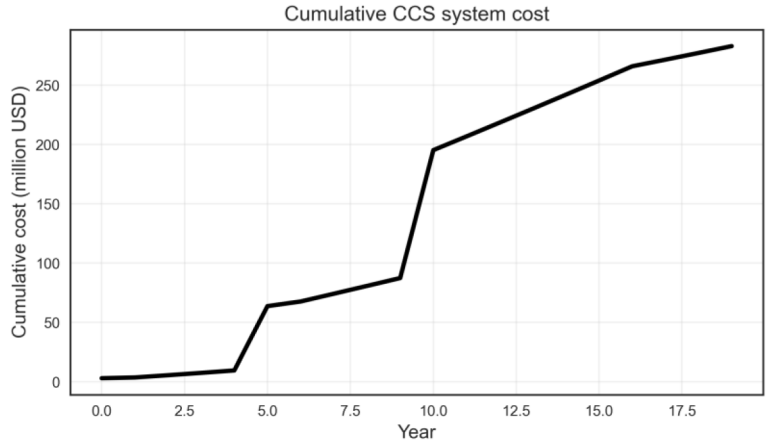


Figure 6—Cumulative system cost

Figure 7 shows the evolution of average reservoir pressure for each candidate storage site. Pressure increases monotonically with cumulative injection. None of the average reservoir pressures (P_r) exceed the fracture pressure limit, confirming that storage capacity is enforced through pressure evolution rather than volumetric constraints.

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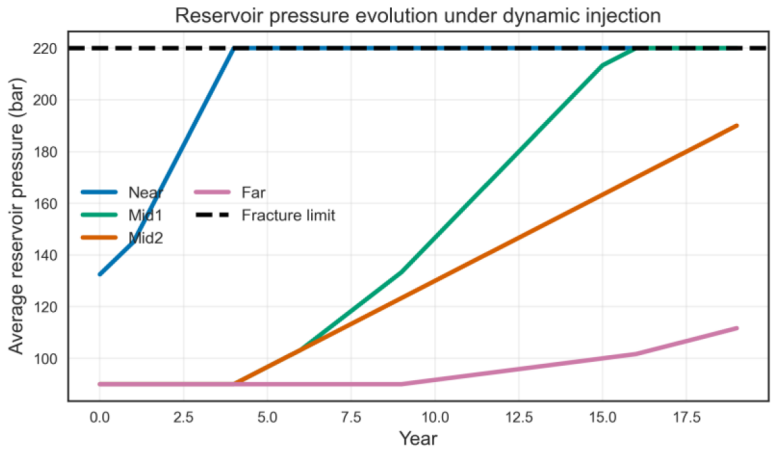


Figure 7—Average reservoir pressure

Figure 8 shows the corresponding bottomhole pressure evolution. Bottomhole pressures were specifically allowed to increase above the fracture limit in this study. Differences between reservoirs are primarily driven by transmissivity and injection rate allocation. The elevated pressure observed in Mid2 is a consequence of its poor injectivity. This result will inform decision-making on whether the reservoir should be excluded, as the required injection pressure is comparatively high.

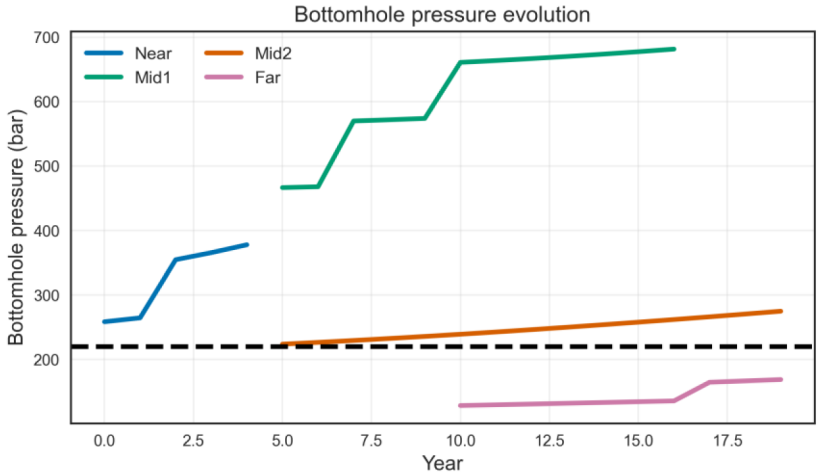


Figure 8—Bottom-hole pressure

Figure 9 presents the evolution of the optimized pipeline internal diameter (ID) for each candidate storage reservoir over the project duration. Pipeline diameters are determined annually through cost minimization and are shown only for years in which a reservoir is actively injected. Near- and mid-field reservoirs require relatively small and weakly time-dependent diameters due to limited injection rates and short transport distances, whereas the far-field reservoir exhibits larger diameters (~20 inch) once it becomes a dominant injection sink.

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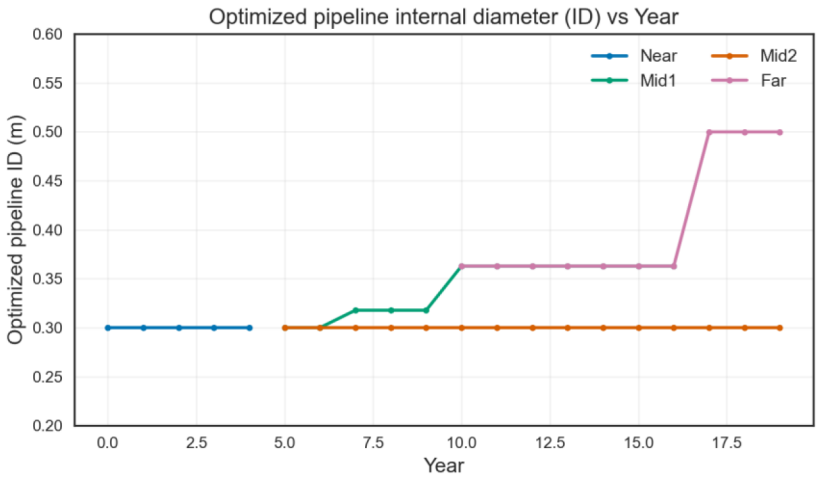


Figure 9—Optimized pipeline internal diameter (ID)

The analysis presented thus far focuses on hydraulic performance without accounting for the global optimization layer using MILP. For example, a manually enforced routing strategy that directs all injection volumes to the Far reservoir (i.e. only use a single 300 km pipeline) results in only a minor increase (0.5 m to 0.55 m) in the maximum required pipeline internal diameter (Figure 10). Doing so would reduce the total cumulative expenditure from approximately USD 230 million (Figure 6) to USD 148 million (Figure 11). While this highlights the significant economic impact of site selection, the specific implementation details of the MILP optimizer, including the comparative performance of nature-inspired stochastic algorithms such as Particle Swarm Optimization (PSO), are omitted from the current scope for brevity.

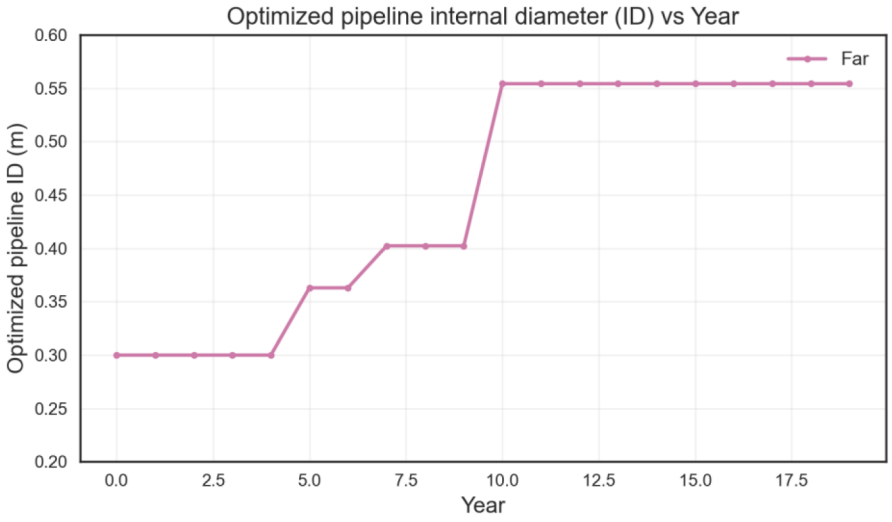


Figure 10—Optimized pipeline internal diameter (ID) using only Far reservoir

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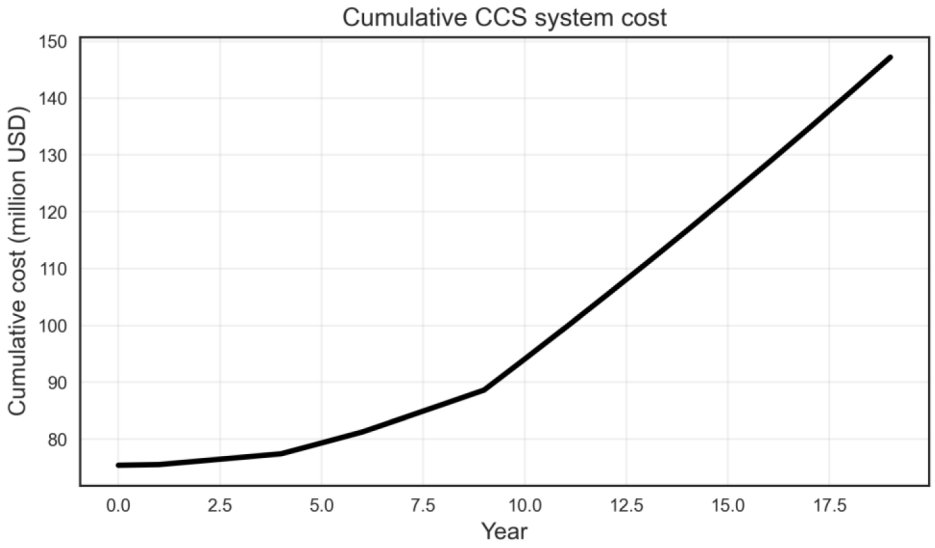


Figure 11—Cumulative system cost using only Far reservoir

Implications for CCS planning

The results demonstrate that CCS network design cannot rely on static cost or distance metrics. Instead, optimal reservoir selection is inherently dynamic, driven by cumulative pressure build-up and injectivity limits. The results show that transporting CO_2 farther to a high-quality reservoir can be economically superior to forcing injection into a nearby but tight formation.

Limitations and Future Extensions

The modelling framework presented in this paper captures the dominant couplings between CO_2 transport, reservoir pressure build-up, and injectivity-controlled well performance. Nevertheless, several simplifying assumptions were made to enable tractable optimisation, and these define clear pathways for future improvement.

Reservoir fracture constraint

In the present method, storage integrity is enforced through a maximum allowable average reservoir pressure, $P_r \leq P_{frac}$. In practice, fracture initiation is governed by the local wellbore pressure rather than the average reservoir pressure. A more physically representative constraint is $P_{bh}(t) \leq P_{frac}$, where P_{bh} is the flowing bottom-hole injection pressure computed from the radial flow equation (Eq. 5). Because $P_{bh} > P_r$, the current formulation is conservative for highly injective reservoirs but may be non-conservative for tighter formations, where large near-well pressure gradients can induce fracture even when the average reservoir pressure remains moderate.

In the present case study, a single injection well is assigned to each pipeline and reservoir. This configuration concentrates the total injection rate into one well, which in some cases leads to elevated bottom-hole pressures (as observed in Figure 8). In realistic CCS developments, injection is typically distributed across multiple wells, allowing total injectivity to be increased while maintaining acceptable wellbore pressures. Incorporating multiple wells would therefore enable the constraint $P_{bh} \leq P_{frac}$ to be enforced more realistically.

A natural extension of the model would replace the tank-model fracture constraint $P_r \leq P_{frac}$ with a wellbore-based constraint of the form $P_{bh}(P_r, m, \dot{k}h) \leq P_{frac}$, thereby coupling injectivity, flow allocation, and fracture risk more directly.

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Hydrostatic pressure effects

The present model neglects hydrostatic pressure effects along pipelines and wellbores. While may acceptable for early-stage screening, hydrostatic pressure can be significant for dense-phase CO_2 transport, particularly in offshore CCS settings with large water depths or deep injection intervals. Future extensions should incorporate hydrostatic pressure contributions to better capture elevation effects on compression requirements and fracture-pressure margins.

Booster stations and multi-segment pipelines

The current model assumes a single compression stage at the capture plant and no intermediate booster stations. For long pipelines, however, the economically optimal configuration often involves multiple compressors to avoid excessive discharge pressure and material limits.

This could be incorporated by splitting the pipeline into segments and introducing booster station variables, with constraints such as:

$$P_{k+1} = P_k - \Delta P_{\text{pipe},k} + \Delta P_{\text{boost},k}$$

The MILP layer could then decide when and where booster stations should be installed.

Network topology

Finally, the current hub-and-spoke configuration assigns one pipeline per reservoir. Future work could include shared trunklines and branching networks, allowing economies of scale and network effects to be captured explicitly.

Conclusions

This study demonstrates that CCS site selection cannot rely on static volumetric capacity alone. Integrating physics-based hydraulic constraints into the early-stage screening process shows that transport distance and reservoir pressure limits may govern the practical feasibility of a storage site more strongly than its theoretical storage volume.

The hydraulic calculations required to evaluate dense-phase CO_2 transport are computationally straightforward to implement and can be applied at an early stage of project development. Deferring such assessments until later project phase is therefore not strictly necessary. Incorporating these calculations during initial screening allows operators to identify critical constraints related to compression requirements and pipeline sizing before undertaking more detailed and resource-intensive analyses, such as transient flow assurance studies.

The framework presented in this paper was developed within the constraints of a short-duration project and is intended as an exploratory, screening-level tool. The modeling components are deliberately simplified and integrated in a pragmatic manner to support rapid assessment rather than comprehensive physical representation. As a result, further work is required to refine the methodology, relax simplifying assumptions, and validate the results against detailed reservoir and flow simulations.

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